COMPSCI 689
Lecture 9: Lagrangian Duality and Kernel SVMs

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An Optimization Problem

An optimization problem in standard form consists of four primary components:

$$\min_{x \in \mathcal{X}} f(\mathbf{x}) \qquad \text{subject to} \begin{cases} c_i(\mathbf{x}) = 0, & i \in \mathcal{E} \\ c_i(\mathbf{x}) \geq 0, & i \in \mathcal{I} \end{cases}$$

- Variables: $\mathbf{x} = [x_1, ..., x_M] \in \mathcal{X}$
- Objective Function: $f : \mathcal{X} \to \mathbb{R}$
- Equality Constraints: $c_i(\mathbf{x}) = 0, i \in \mathcal{E}$
- Inequality Constraints: $c_i(\mathbf{x}) \geq 0, i \in \mathcal{I}$

SVC Primal QP

$$\hat{\theta} = \arg\min_{\theta, \epsilon_{1:N}} \mathbf{w}^T \mathbf{w} + C \sum_{n=1}^{N} \epsilon_n$$

$$\text{s.t. } \forall n \; y_n g_\theta(\mathbf{x}_n) \geq 1 - \epsilon_n$$

$$\forall n \; \epsilon_n \geq 0$$

The Lagrangian

- The fundamental tool for analyzing constrained optimization problems is the Lagrangian function.
- Given an objective function $f(\mathbf{x})$ and a set of equality and inequality constraint functions $c_i(\mathbf{x}) = 0$ for $i \in \mathcal{E}$ and $c_i(\mathbf{x}) \geq 0$ for $i \in \mathcal{I}$, the Lagrangian is the function:

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \sum_{i \in \mathcal{I}} \lambda_i c_i(\mathbf{x}) - \sum_{i \in \mathcal{E}} \lambda_i c_i(\mathbf{x})$$

- The new variables λ_i are referred to as *Lagrange Multipliers*.
- The Karush-Kuhn-Tucker (KKT) conditions specify the necessary first-order conditions for $\mathbf{x}_*$ to be the constrained minimizer of $f(\mathbf{x})$ in terms of the Lagrangian $\mathcal{L}(\mathbf{x}, \lambda)$.

The KKT Conditions

- Suppose that $\mathbf{x}_*$ is a local minima of a constrained optimization problem with Lagrangian $\mathcal{L}(\mathbf{x}, \lambda)$ and constraints $c_i(\mathbf{x}) = 0$ for $i \in \mathcal{E}$ and $c_i(\mathbf{x}) \geq 0$ for $i \in \mathcal{I}$. Then there exists a λ_* such that:

$$
\begin{aligned}
\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}_*, \lambda_*) &= 0 \\
c_i(\mathbf{x}_*) &= 0 \text{ for all } i \in \mathcal{E} \\
c_i(\mathbf{x}_*) &\geq 0 \text{ for all } i \in \mathcal{I} \\
\lambda_{i*} &\geq 0 \text{ for all } i \in \mathcal{I} \\
\lambda_{i*} c_i(\mathbf{x}_*) &= 0 \text{ for all } i \in \mathcal{I}
\end{aligned}
$$

- For convex problems, if $(\mathbf{x}_*, \lambda_*)$ satisfy the KKT conditions, then $\mathbf{x}_*$ is the global constrained minimizer.

The Lagrange Dual

- Given the Lagrangian for an objective $f(x)$ and constraint functions $c_i(\mathbf{x}) \geq 0$ and $c_i(\mathbf{x}) = 0$:

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \sum_{i \in \mathcal{E}} \lambda_i c_i(\mathbf{x}) - \sum_{i \in \mathcal{I}} \lambda_i c_i(\mathbf{x})$$

- The Lagrange dual function is given by:

$$q(\lambda) = \min_{\mathbf{x} \in \mathcal{X}} \mathcal{L}(\mathbf{x}, \lambda)$$

- This function is subject to the constraints $\lambda_i \geq 0$ for $i \in \mathcal{I}$.

Lagrange Dual Optimization

1. Given $f(\mathbf{x})$ and constraint functions $c_i(\mathbf{x})$, form the Lagrangian $\mathcal{L}(\mathbf{x}, \lambda)$.
2. Derive the gradient $\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \lambda)$.
3. Form the stationary equations $\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \lambda) = 0$.
4. The stationary equations will yield relationships between primary parameters and Lagrange multipliers that hold when $\mathbf{x}_* = \arg\min_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \lambda)$.
5. These relationships can be used to eliminate the primary parameters from the Lagrangian, yielding $q(\lambda) = \mathcal{L}(\mathbf{x}_*, \lambda)$ and possibly some additional equality constraints.
6. We then solve the optimization problem $\lambda_* = \arg\max_{\lambda} q(\lambda)$ subject to $\lambda_i \geq 0$ and any other constraints.

Properties of the Lagrange Dual

- For strongly convex functions with convex constraints, a property called *strong duality* holds:

$$\min_{\mathbf{x}\in\mathcal{X}} f(\mathbf{x}) = \max_{\lambda\in\Lambda} q(\lambda)$$

- Further, we can typically use the constrained dual maximizer λ_* to construct the constrained primal minimizer $\mathbf{x}_*$.
- This is useful when it is preferable to solve the dual optimization problem $\arg\max_\lambda q(\lambda)$ either because the primal can not be solved analytically, because it is computationally more efficient to solve the dual, or because the dual enables functionality not accessible via the primal.

Problem and Lagrangian

We are tasked with solving the following optimization problem under the assumption that x is a column vector and A is strictly positive definite and symmetric matrix.

$$\min_x \; x^T A x \quad \text{subject to} \quad x^T b + c \geq 0$$

We introduce the Lagrange multiplier $\lambda \geq 0$ for the inequality constraint $x^T b + c \geq 0$. The Lagrangian is:

$$\mathcal{L}(x, \lambda) = x^T A x - \lambda(x^T b + c)$$

Minimization wrt x

To find the Lagrange dual function, we minimize the Lagrangian with respect to x. First, we take the gradient of the Lagrangian with respect to x and set it equal to zero:

$$\nabla_x \mathcal{L}(x, \lambda) = \nabla_x (x^T A x - \lambda(x^T b + c)) = 2Ax - \lambda b = 0.$$

Solving for x, we get:

$$x_* = \frac{\lambda}{2} A^{-1} b,$$

Construction of the Dual

Now, we substitute $x_* = \frac{\lambda}{2}A^{-1}b$ into the Lagrangian to find the dual function $q(\lambda)$:

$$
\begin{aligned}
q(\lambda) &= x_*^T A x_* - \lambda(x_*^T b + c) \\
&= \frac{\lambda^2}{4} b^T A^{-1} b - \lambda\left(\frac{\lambda}{2} b^T A^{-1} b + c\right) \\
&= \frac{\lambda^2}{4} b^T A^{-1} b - \frac{\lambda^2}{2} b^T A^{-1} b - \lambda c \\
&= -\frac{\lambda^2}{4} b^T A^{-1} b - \lambda c
\end{aligned}
$$

Maximization of the Dual

To maximize the dual function $q(\lambda)$, we take the derivative with respect to λ and set it equal to zero:

$$\begin{aligned}\frac{d}{d\lambda}q(\lambda) &= \frac{d}{d\lambda}(-\frac{\lambda^2}{4}b^TA^{-1}b - \lambda c)\\ &= -\frac{1}{2}\lambda b^TA^{-1}b - c = 0\end{aligned}$$

Solving for λ, we get:

$$\lambda_* = -\frac{2c}{b^TA^{-1}b}$$

Since $\lambda \geq 0$, we have the following result:

$$\lambda_* = \max\left(0, -\frac{2c}{b^TA^{-1}b}\right).$$

Thus, if $c \leq 0$, the optimal $\lambda_* = -\frac{2c}{b^TA^{-1}b}$. If $c > 0$, the optimal $\lambda_* = 0$.

Finding the Optimal x

Using the expression $x_* = \frac{\lambda}{2}A^{-1}b$, we substitute the optimal λ^*:

For $c \leq 0$ we have $\lambda_* = -\frac{2c}{b^T A^{-1} b}$ and the optimal x is:

$$x_* = \frac{\lambda_*}{2}A^{-1}b = -\frac{cA^{-1}b}{b^T A^{-1} b}$$

For $c > 0$ we have $\lambda_* = 0$ and the optimal x is:

$$x_* = \frac{\lambda_*}{2}A^{-1}b = 0.$$

The SVC Dual Optimization Problem

- By applying Lagrangian duality to the SVC primal QP, we can derive the The SVC dual optimization problem below where we have used α_n as the Lagrange multipliers.

$$\begin{aligned}
\arg\max_{\boldsymbol{\alpha}} \sum_{n=1}^{N} \alpha_n - \frac{1}{2}\sum_{n=1}^{N}\sum_{m=1}^{N} y_n y_m \alpha_n \alpha_m \mathbf{x}_n \mathbf{x}_m^T \\
\text{s.t. } \forall n \;\; 0 \leq \alpha_n \leq C \\
\sum_{n=1}^{N} \alpha_n y_n = 0
\end{aligned}$$

- This is a concave objective with box and linear equality constraints.

SVC Dual Predictions

- Given the optimal value of $\boldsymbol{\alpha}$, the optimal weight vector can be computed as $\mathbf{w}_* = \sum_{n=1}^{N} y_n \alpha_n \mathbf{x}_n^T$ leading to the prediction function:

$$f_{svm}(\mathbf{x}) = \text{sign}\left(\sum_{n=1}^{N} y_n \alpha_n \mathbf{x}\mathbf{x}_n^T + b\right)$$

- There are a number of ways to derive the optimal b from the KKT conditions and an optimal $\boldsymbol{\alpha}$. One approach is to use the equation below where $\mathcal{S}$ is the set of data points satisfying $0 < \alpha_n < C$:

$$b = \frac{1}{|\mathcal{S}|} \sum_{n \in \mathcal{S}} \left(y_n - \sum_{m \in \mathcal{S}} y_m \alpha_m \mathbf{x}_m \mathbf{x}_n^T\right)$$

Support Vector Property

- The above results show that only the data points for which $\alpha_n > 0$ actually participate in defining the learned model and making predictions.
- The data points for which $\alpha_n > 0$ are called *support vectors*. For the linearly separable case, these are the data cases on the margin. For the non-separable case, they are the data points on, within and on the wrong side of the margin.

The SVC Dual Optimization Problem

- We note that the dual only depends on the inputs via inner products $\mathbf{x}_n\mathbf{x}_m^T$. We can thus write the problem as shown below where $\mathbf{K}_{nm} = \mathbf{x}_n\mathbf{x}_m^T$.

$$\begin{aligned}
&\arg\max_{\boldsymbol{\alpha}} \sum_{n=1}^{N} \alpha_n - \frac{1}{2}\sum_{n=1}^{N}\sum_{m=1}^{N} y_n y_m \alpha_n \alpha_m \mathbf{K}_{nm} \\
&\text{s.t. } \forall n \;\; 0 \leq \alpha_n \leq C \\
&\sum_{n=1}^{N} \alpha_n y_n = 0
\end{aligned}$$

- The matrix $\mathbf{K}$ that appears in the objective contains the inner products between all pairs of training data vectors. The matrix $\mathbf{K}$ is thus positive semi-definite and the optimization problem is maximizing a concave function.

Basis Expansion

- One of the important applications of the SVC dual problem is constructing non-linear classifiers.
- Recall that we can make a linear model non-linear by introducing a basis function expansion $\phi(\mathbf{x})$.
- Plugging this into the dual, we obtain the learning problem shown below where the only change is $\mathbf{K}_{nm} = \phi(\mathbf{x}_n)\phi(\mathbf{x}_m)^T$.

$$\begin{aligned}
&\arg\max_{\boldsymbol{\alpha}} \sum_{n=1}^{N} \alpha_n - \frac{1}{2}\sum_{n=1}^{N}\sum_{m=1}^{N} y_n y_m \alpha_n \alpha_m \mathbf{K}_{nm} \\
&\text{s.t. } \forall n \;\; 0 \leq \alpha_n \leq C \\
&\quad \sum_{n=1}^{N} \alpha_n y_n = 0
\end{aligned}$$

Kernels

- Interestingly, the **only** requirement for the optimization problem to be well-defined is for the matrix $\mathbf{K}$ to be positive semi-definite.
- An alternative to using an explicit basis expansion is to use a kernel function $\mathcal{K}(\mathbf{x}, \mathbf{x}')$ that is guaranteed to generate a positive definite matrix $\mathbf{K}$ given any collection of data vectors $\mathbf{x}_n$. Such a kernel function is referred to as a *Mercer* kernel.
- In the learning problem, instead of setting $\mathbf{K}_{nm} = \phi(\mathbf{x}_n)\phi(\mathbf{x}_m)^T$, we set $\mathbf{K}_{nm} = \mathcal{K}(\mathbf{x}_n, \mathbf{x}_m)$.
- The prediction function becomes:

$$f_{svm}(\mathbf{x}) = \left(\sum_{n=1}^{N} y_n \alpha_n \mathcal{K}(\mathbf{x}_n, \mathbf{x}) + b\right)$$

Example: Kernels

- Linear Kernel: $\mathcal{K}(\mathbf{x}', \mathbf{x}) = \mathbf{x}'\mathbf{x}^T$
- Polynomial Kernel: $\mathcal{K}(\mathbf{x}', \mathbf{x}) = (1 + \mathbf{x}'\mathbf{x}^T)^B$
- Exponential (RBF) Kernel: $\mathcal{K}(\mathbf{x}', \mathbf{x}) = a \exp\left(-b\|\mathbf{x}' - \mathbf{x}\|_2^2 \right)$
- ... and many more kernel functions, including for inputs that are not real-valued.

Properties of Duals and Kernels

- The optimization problem remains concave when using kernels, but the decision function can be complex and non-linear.
- When there existis a kernel implementing a given basis expansion, the dual formulation results in a QP that scales with N while the primal results in a QP that scales with the size of the basis expansion.
- There exist kernel functions (like the RBF kernel) that do not correspond to finite-dimensional basis expansions that can be used exactly in the dual, but must be approximated to use in the primal.

Generalizing Kernelization

- Any convex supervised learning optimization problem with convex (or no constraints) has a similar dual to SVC and can usually be kernelized as well (linear regression, SVR, logistic regression, GLMs, etc.).
- However, only SVC and SVR have a direct sparsity property where we typically expect many dual variables to be zero.

The SVC Primal Optimization Problem

- Recall that the constrained version of the SVC optimization problem has the form shown below.

$$\begin{aligned} \arg\min_{\mathbf{w},b,\epsilon} \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{n=1}^{N}\epsilon_n \\ \text{s.t. } \forall n \;\; y_n(\mathbf{x}_n\mathbf{w}+b) \geq 1-\epsilon_n \\ \forall n \;\; \epsilon_n \geq 0. \end{aligned}$$

- This problem has a convex objective with convex constraints. We will derive its Lagrange dual.

Lagrangian

$$
\begin{aligned}
\arg\min_{\mathbf{w},b,\epsilon} \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{n=1}^{N}\epsilon_n \\
\text{s.t. } \forall n \;\; y_n(\mathbf{x}_n\mathbf{w} + b) \geq 1 - \epsilon_n \\
\forall n \;\; \epsilon_n \geq 0.
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{L}(\mathbf{w}, b, \epsilon, \alpha, \beta) =& \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{n=1}^{N}\epsilon_n \; - \sum_{n=1}^{N}\beta_n\epsilon_n \\
& - \sum_{n=1}^{N}\alpha_n(y_n(\mathbf{x}_n\mathbf{w} + b) + \epsilon_n - 1) \\
& \text{s.t. } \forall n \;\; \alpha_n \geq 0, \beta_n \geq 0
\end{aligned}
$$

Minimizing wrt w

$$\mathcal{L}(\mathbf{w}, b, \epsilon, \alpha, \beta) = \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{n=1}^{N}\epsilon_n - \sum_{n=1}^{N}\beta_n\epsilon_n - \sum_{n=1}^{N}\alpha_n(y_n(\mathbf{x}_n\mathbf{w} + b) + \epsilon_n - 1)$$
$$\text{s.t. } \forall n \ \ \alpha_n \geq 0, \beta_n \geq 0$$

$$\nabla_{\mathbf{w}}\mathcal{L}(\mathbf{w}, b, \epsilon, \alpha, \beta) = \mathbf{w} - \sum_{n=1}^{N}\alpha_n y_n \mathbf{x}_n^T = 0$$
$$\mathbf{w} = \sum_{n=1}^{N}\alpha_n y_n \mathbf{x}_n^T$$

Minimizing wrt b

$$\mathcal{L}(\mathbf{w}, b, \epsilon, \alpha, \beta) = \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{n=1}^{N} \epsilon_n - \sum_{n=1}^{N} \beta_n \epsilon_n - \sum_{n=1}^{N} \alpha_n (y_n(\mathbf{x}_n \mathbf{w} + b) + \epsilon_n - 1)$$

$$\text{s.t. } \forall n \;\; \alpha_n \geq 0, \beta_n \geq 0$$

$$\nabla_b \mathcal{L}(\mathbf{w}, b, \epsilon, \alpha, \beta) = -\sum_{n=1}^{N} \alpha_n y_n = 0$$

$$\sum_{n=1}^{N} \alpha_n y_n = 0$$

Minimizing wrt ϵ

$$\begin{aligned}\mathcal{L}(\mathbf{w}, b, \epsilon, \alpha, \beta) =& \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{n=1}^{N}\epsilon_n \; - \sum_{n=1}^{N}\beta_n\epsilon_n \\ & - \sum_{n=1}^{N}\alpha_n(y_n(\mathbf{x}_n\mathbf{w} + b) + \epsilon_n - 1) \\ & \text{s.t. } \forall n \;\; \alpha_n \geq 0, \beta_n \geq 0\end{aligned}$$

$$\begin{aligned}\nabla_{\epsilon_n}\mathcal{L}(\mathbf{w}, b, \epsilon, \alpha, \beta) = C - \beta_n - \alpha_n = 0 \\ \alpha_n + \beta_n = C\end{aligned}$$

Substituting into Lagrangian

$$
\begin{aligned}
\mathcal{L}(\mathbf{w}, b, \epsilon, \alpha, \beta) = & \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{n=1}^{N} \epsilon_n \; - \sum_{n=1}^{N} \beta_n \epsilon_n \\
& - \sum_{n=1}^{N} \alpha_n (y_n(\mathbf{x}_n \mathbf{w} + b) + \epsilon_n - 1)
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{L}(\mathbf{w}, b, \epsilon, \alpha, \beta) = & \frac{1}{2}\| \sum_{m=1}^{N} \alpha_m y_m \mathbf{x}_m^T \|_2^2 + C\sum_{n=1}^{N} \epsilon_n \; - \sum_{n=1}^{N} \beta_n \epsilon_n \\
& - \sum_{n=1}^{N} \alpha_n (y_n(\mathbf{x}_n (\sum_{m=1}^{N} \alpha_m y_m \mathbf{x}_m^T) + b) + \epsilon_n - 1)
\end{aligned}
$$

Expanding Terms

$$\mathcal{L}(\mathbf{w}, b, \epsilon, \alpha, \beta) = \frac{1}{2} \| \sum_{m=1}^{N} \alpha_m y_m \mathbf{x}_m^T \|_2^2 + C \sum_{n=1}^{N} \epsilon_n - \sum_{n=1}^{N} \beta_n \epsilon_n$$
$$- \sum_{n=1}^{N} \alpha_n (y_n (\mathbf{x}_n (\sum_{m=1}^{N} \alpha_m y_m \mathbf{x}_m^T) + b) + \epsilon_n - 1)$$

$$\mathcal{L}(\mathbf{w}, b, \epsilon, \alpha, \beta) = \frac{1}{2} \sum_{n=1}^{N} \sum_{m=1}^{N} \alpha_n \alpha_m y_n y_m \mathbf{x}_n \mathbf{x}_m^T + C \sum_{n=1}^{N} \epsilon_n - \sum_{n=1}^{N} \beta_n \epsilon_n$$
$$- \sum_{n=1}^{N} \sum_{m=1}^{N} \alpha_n \alpha_m y_n y_m \mathbf{x}_n \mathbf{x}_m^T - \sum_{n=1}^{N} \alpha_n (y_n b + \epsilon_n - 1)$$

Simplifying I

$$
\begin{aligned}
\mathcal{L}(\mathbf{w}, b, \epsilon, \alpha, \beta) = & \frac{1}{2}\sum_{n=1}^{N}\sum_{m=1}^{N}\alpha_n\alpha_m y_n y_m \mathbf{x}_n\mathbf{x}_m^T + C\sum_{n=1}^{N}\epsilon_n - \sum_{n=1}^{N}\beta_n\epsilon_n \\
& - \sum_{n=1}^{N}\sum_{m=1}^{N}\alpha_n\alpha_m y_n y_m \mathbf{x}_n\mathbf{x}_m^T - \sum_{n=1}^{N}\alpha_n(y_n b + \epsilon_n - 1)
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{L}(\mathbf{w}, b, \epsilon, \alpha, \beta) = & -\frac{1}{2}\sum_{n=1}^{N}\sum_{m=1}^{N}\alpha_n\alpha_m y_n y_m \mathbf{x}_n\mathbf{x}_m^T + \sum_{n=1}^{N}(\alpha_n + \beta_n)\epsilon_n \\
& - \sum_{n=1}^{N}\beta_n\epsilon_n - \sum_{n=1}^{N}\alpha_n(y_n b + \epsilon_n - 1)
\end{aligned}
$$

Simplifying II

$$\mathcal{L}(\mathbf{w}, b, \epsilon, \alpha, \beta) = -\frac{1}{2}\sum_{n=1}^{N}\sum_{m=1}^{N}\alpha_n\alpha_m y_n y_m \mathbf{x}_n\mathbf{x}_m^T + \sum_{n=1}^{N}(\alpha_n + \beta_n)\epsilon_n - \sum_{n=1}^{N}\beta_n\epsilon_n - \sum_{n=1}^{N}\alpha_n(y_n b + \epsilon_n - 1)$$

$$\mathcal{L}(\mathbf{w}, b, \epsilon, \alpha, \beta) = -\frac{1}{2}\sum_{n=1}^{N}\sum_{m=1}^{N}\alpha_n\alpha_m y_n y_m \mathbf{x}_n\mathbf{x}_m^T + \sum_{n=1}^{N}\alpha_n$$

Constraint Simplification

$$\begin{aligned} \alpha_n &\geq 0 \\ \beta_n &\geq 0 \\ \sum_{n=1}^{N} \alpha_n y_n &= 0 \\ \alpha_n + \beta_n &= C \end{aligned}$$

$$\begin{aligned} \sum_{n=1}^{N} \alpha_n y_n &= 0 \\ 0 \leq \alpha_n &\leq C \end{aligned}$$